

Mathematical aspects and trajectory factors

1. Analysis of Technical Reference Manual (PDF Review)

Summary & Core Concepts

The provided document, **“Ballistic Missile Technical Reference Guide: Structural Anatomy, Component Visual Profiles, and Technical Specifications for Selected Chinese (PLARF) and Pakistani (ASFC) Missile Systems”**, serves as a high-level unclassified technical intelligence reference sheet. It outlines the structural anatomy, component breakdown, and macro specifications across 16 ballistic missile platforms ranging from Tactical/Sub-Strategic systems to Heavy Intercontinental Ballistic Missiles (ICBMs)

2. Mathematical Optimization Formulation

Trajectory & Candidate Site Models

Let a synthetic target trajectory $i \in \{1, \dots, I\}$ be represented as a time-indexed state sequence over discretization $k \in \{1, \dots, K_i\}$:

$$T_i = \{(t_k, \phi_k, \lambda_k, h_k, v_k, \gamma_k)\}_{k=1}^{K_i}$$

where t_k is timestamp, (ϕ_k, λ_k) are latitude and longitude, h_k is geometric altitude, v_k is scalar velocity, and γ_k is the flight-path angle.

Let a candidate interceptor site $j \in \{1, \dots, J\}$ be defined as:

$$S_j = (P_j, n_j, c_j, q_j)$$

where P_j is the geospatial polygon boundary, $n_j \in \mathbb{Z}^+$ is total abstract interceptor inventory capacity, $c_j \in \mathbb{R}^+$ is the normalized deployment cost, and $q_j \in [0, 1]$ is a composite site quality index combining communications, terrain, and sensor support.

Engagement & Probabilistic Matrices

Define the binary kinematic feasibility matrix $a_{ij} \in \{0, 1\}$ as:

$$a_{ij} = \begin{cases} 1, & \text{if candidate site } j \text{ satisfies kinematic, temporal, and sector constraints for target } i \\ 0, & \text{otherwise} \end{cases}$$

The overall engagement probability $p_{ij} \in [0, 1]$ is the multiplicative product of independent success probabilities across system layers:

$$p_{ij} = p_{\text{detect},ij} \cdot p_{\text{track},ij} \cdot p_{\text{timing},ij} \cdot p_{\text{engage},ij} \cdot p_{\text{kill},ij} \cdot g_{ij}$$

If $a_{ij} = 0$ or $g_{ij} = 0$, then $p_{ij} = 0$.

Decision Variables

- $x_j \in \{0, 1\}$: Binary decision variable indicating whether candidate site j is deployed ($x_j = 1$) or not ($x_j = 0$).
- $z_{ij} \in \mathbb{Z}_{\geq 0}$: Integer decision variable representing the number of abstract interceptors assigned from site j to target i .

Target Protection Probability

Under the assumption of independent interceptor engagements, if z_{ij} interceptors are allocated from site j to target i , the total cumulative probability of target interception P_i is given by:

$$P_i = 1 - \prod_{j=1}^J (1 - p_{ij})^{z_{ij}}$$

Mixed-Integer Linear Optimization Model (Linearized Form)

To optimize via MILP solvers (e.g., PuLP, OR-Tools), we linearize the non-linear objective P_i using standard logarithmic transformation or piecewise linear approximation.

Objective Function

$$\max_{x,z} \sum_{i=1}^I w_i P_i - \alpha \sum_{j=1}^J c_j x_j - \beta \sum_{i=1}^I \max(0, R_i - P_i)$$

where w_i is target priority weight, α is cost penalty coefficient, R_i is required protection probability threshold, and β is penalty weight for failing to meet R_i

Constraints

1. Site Allocation Capacity:

$$\sum_{i=1}^I z_{ij} \leq n_j x_j \quad \forall j \in \{1, \dots, J\}$$

2. Feasibility Alignment:

$$z_{ij} \leq n_j a_{ij} \quad \forall i, j$$

3. Simultaneous Engagement Limit:

$$\sum_{i \in \Omega(t)} z_{ij} \leq M_j \quad \forall j, \forall t$$

where $\Omega(t)$ is the set of targets active in time window t , and M_j is the maximum simultaneous engagement capacity of site j .

4. Binary & Integer Bounds:

$$x_j \in \{0, 1\} \quad \forall j, \quad z_{ij} \in \{0, 1, \dots, n_j\} \quad \forall i, j$$

3. Safe Synthetic Dataset

A 100 km × 100 km bounding box is set up in a generic planar coordinate frame centered at local projected origin (0, 0) km.

Candidate Sites Table (CSV)

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site_id , poly_wkt , capacity_n , cost_c , quality_q, layer
S01,"POLYGON((10 10, 15 10, 15 15, 10 15, 10 10))" ,8,3.5,0.85,Endo
S02,"POLYGON((25 40, 30 40, 30 45, 25 45, 25 40))" ,12,6.0,0.92,Dual
S03,"POLYGON((50 15, 55 15, 55 20, 50 20, 50 15))" ,6,2.0,0.70,Endo
S04,"POLYGON((70 70, 75 70, 75 75, 70 75, 70 70))" ,10,5.5,0.88,Exo
S05,"POLYGON((15 80, 20 80, 20 85, 15 85, 15 80))" ,8,4.0,0.78,Endo
S06,"POLYGON((80 20, 85 20, 85 25, 80 25, 80 20))" ,14,7.5,0.95,Dual
S07,"POLYGON((40 50, 45 50, 45 55, 40 55, 40 50))" ,10,5.0,0.82,Endo
S08,"POLYGON((60 30, 65 30, 65 35, 60 35, 60 30))" ,6,2.5,0.65,Endo
```

Target Trajectories Table (CSV)

target id , scenario type , t_start , t_end, start_x_km, start_y_km, end_x_km,end_y_km, apogee_km, speed_km_s, req_protection

T01, Short-Range, 0 ,45 ,5 ,5 ,35 ,35 ,25 ,1.2 ,0.80

T02 , Short-Range, 5, 50, 10, 20, 40, 50, 30, 1.4, 0.80

T03, Short-Range, 10, 55 ,5 ,70 ,35 ,80 ,28,1.3,0.75

T04, Medium-Range,0, 90, 0, 10,70, 80, 65, 2.5, 0.85

T05, Medium-Range,15, 105, 10, 0, 80, 90,70, 2.8, 0.85

T06, Medium-Range, 20, 110, 20, 100, 80, 10, 60, 2.4, 0.85

T07, Medium-Range, 10, 95, 90, 5, 10, 85, 68, 2.6, 0.85

T08, Long-Range, 0, 180, 0, 0, 95, 95, 140, 4.5, 0.90

T09, Long-Range, 30, 210, 100, 0, 5, 90, 150, 4.8, 0.90

T10, Long-Range, 20, 195, 0, 50, 90, 50, 135, 4.2, 0.90

software Architecture

Algorithm & Pseudocode

ALGORITHM: Defense Site Selection & Resource Allocation

1. Find Target Positions

Calculate the midpoint and peak height (apogee) of each incoming missile's flight path.

2. Check Feasibility & Accuracy

-> Measure the distance from every interceptor base to each missile's midpoint.

- > If the missile is **within range**, mark it as reachable and calculate the hit chance based on distance and base quality.

-> If the missile is **too far**, mark the hit chance as 0%.

3. Pick the Best Defense Plan

Use an optimization solver to pick which bases to open and how many defense missiles to fire so that:

-> High-priority targets get the most protection.

-> Total money spent opening bases is kept as low as possible.

-> No base fires more missiles than it holds in stock

Simple Pseudocode

FOR EACH target missile:

start point : Find the midpoint of its path

FOR EACH defense site:

Calculate distance to missile midpoint

IF distance is within max range THEN:

1. Mark site as FEASIBLE

2. Calculate HIT CHANCE based on distance and site quality

ELSE:

Mark site as NOT FEASIBLE

Set HIT CHANCE to 0

END IF

END FOR

END FOR

CHOOSE defense sites to open AND allocate missiles TO:

1. Maximize overall protection for all targets

2. Minimize total cost of opening defense sites

3. Keep site missile limits respected

OUTPUT chosen sites, missile assignments, and protection levels.